

Sardar Vallabhbhai National Institute of Technology, Surat
Computer Science and Engineering Department
M.Tech(CSE/DS)-I - Sem-I - 2026-27
Computer Vision and Image Processing (CSCS111/CSDS119)
Lab Assignment 6: Image Segmentation

PART A:

1. Take a photograph of yourself holding an object of your choice (e.g., a book, bottle, mobile phone, cup, or any other object). Use the image to explore different image segmentation techniques.

- Detect prominent points and lines in the image using suitable point and directional line detection masks.
- Detect the boundaries of yourself and the object using suitable edge detection operators such as Roberts, Prewitt, Sobel, and Laplacian.
- Segment yourself and/or the object from the background using thresholding-based segmentation. Apply global thresholding and Otsu's thresholding and compare the results.
- Apply a suitable region-based segmentation technique and investigate how the choice of seed point or similarity criterion affects the segmented region.
- Display the original image and the results obtained from the different segmentation techniques. Write your observations on which technique is able to separate the object from the background more effectively and why.

Note: For reading and writing images, you may use built-in functions from OpenCV or any other library you use.

PART B:

Can you make your object disappear?

Using the same photograph, suppose you want to create an image in which only the object in your hand is retained and everything else is removed. However, the object has a similar intensity or colour to part of the background.

Design a segmentation pipeline using one or more techniques studied in this lab. You are free to combine point detection, line detection, edge detection, thresholding, or region-based methods. Compare your result with a single segmentation technique and explain why your proposed combination works better or where it fails.